Quantum Games and Nonlocality

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Namit Anand

1111033

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Under the guidance of

Dr. Colin Benjamin

National Institute of Science Education and Research

Bhubaneswar - 751005

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Chapter 1

Classical Games

1.1 Introduction

Game theory is a branch of Mathematics which deals with situations in which decision makers(players) interact.
When one encounters the theory of games, some naturally arising questions are:
What is a game ?
A game is a form of **competitive** activity or sport played according to **rules**.
Why should we study game theory ?
Simply because it is there in nature and like all phenomena in nature, it too is very interesting.
Survival of the fittest (Natural Selection) is one of the oldest games played by different species of organisms.
Just to give an idea of the range of situations to which game theory can be applied: firms competing for business, political candidates competing for votes, jury members deciding on a verdict, animals fighting over prey, bidders competing in an auction, the evolution of siblings behavior towards each other, competing experts incentives to provide correct diagnoses, legislators voting behavior under pressure from interest groups, and the role of threats and punishment in long-term relationships.

Figure 1.1: Chess - A combinatorial game

1.2 Basic Definitions

1.2.1 Theory of Rational Choice

The theory of rational choice is a component of many models in game theory. Briefly, this theory is that a decision-maker chooses the best action according to her preferences, among all the actions available to her. No qualitative restriction is placed on the decision-makers preferences; her rationality lies in the consistency of her decisions when faced with different sets of available actions, not in the nature of her likes and dislikes.

1.2.2 Actions

The theory is based on a model with two components: a set A consisting of all the actions that, under some circumstances, are available to the decision-maker, and a specification of the decision-makers preferences. In any given situation the decision-maker is faced with a subset1 of A, from which she must choose a single element. The decision-maker knows this subset of available choices, and takes it as given; in particular, the subset is not influenced by the decision-makers preferences. The set A could, for example, be the

set of bundles of goods that the decision-maker can possibly consume; given her income at any time, she is restricted to choose from the subset of A containing the bundles she can afford.

1.2.3 Preferences and payoff functions

As to preferences, we assume that the decision-maker, when presented with any pair of actions, knows which of the pair she prefers, or knows that she regards both actions as equally desirable (is indifferent between the actions). We can represent the preferences by a payoff function, which associates a number with each action in such a way that actions with higher numbers are preferred.
More precisely, the payoff function u represents a decision-makers preferences if, for any actions a in A and b in A,
u(a) > u(b) if and only if the decision-maker prefers a to b.

1.3 Strategic Theory

A **STRATEGIC GAME** is a model of interacting decision-makers. In recognition of the interaction, we refer to the decision-makers as players. Each player has a set of possible actions. The model captures interaction between the players by allowing each player to be affected by the actions of all players, not only her own action. Specifically, each player has preferences about the action profile - the list of all the players actions.

1.3.1 Defintion

A strategic game (with ordinal preferences) consists of:

- a set of players
- for each player, a set of actions
- for each player, preferences over the set of action profiles.

As in the model of rational choice by a single decision-maker (Section 1.2), it is frequently convenient to specify the players preferences by giving payoff functions that represent them.
Time is absent from the model. The idea is that each player chooses her action once and for all, and the players choose their actions simultaneously in the sense that no player is informed, when she chooses her action, of the action chosen by any other player. (For this reason, a strategic game is sometimes referred to as a simultaneous move game.)

1.3.2 Example: Prisoner's Dilemma

Two thieves plan to rob an electronics store. As they approach the back-door, the police arrest them for trespassing. The cops suspect that the pair had planned to break in but lack the evidence to support such an accusation. They therefore require a confession to charge the suspects with the greater crime.
Having studied game theory in college, the interrogator throws them into the prisoners dilemma. He privately sequesters both robbers and tells them the following:

We are currently charging you with trespassing, which calls for a one month jail sentence. I know you were planning on robbing the store, but right now I cannot prove itI need your testimony. In exchange for your cooperation, I will dismiss your trespassing charge, and your partner will be charged to the fullest extent of the lawa twelve month jail sentence. I am offering your partner the same deal. If both of you confess, your individual testimony is no longer as valuable, and your jail sentence will be eight months each.
If both criminals are self-interested and only care about minimizing their jail time, should they take the interrogators deal?
This situation may be modeled as a strategic game:
Players: The two suspects.
Actions: Each players set of actions is Keep Quiet, Confess.

Solving the game

The game can be solved in several ways but one of the simplest ways to condense all that we know, is to put it in a matrix form, also known as the **normal form** representation of a game:

The first entry is the payoff of Player 2 and the second is that of Player 1.

	Keep Quiet	Confess
Keep Quiet	-1,-1	-12,0
Confess	0,-12	-8,-8

To find out the optimal strategy, we consider the action of Player 1(say) arbitrary and look at Player 2's payoff.
Case 1:
Player 1 chooses KEEP QUIET then

	Keep Quiet
Keep Quiet	-1,-1
Confess	0,-12

One can clearly see that Confess is a better option for Player 2 since $0 > -1$
Case 1:

Player 1 chooses CONFESS then

	Confess
Keep Quiet	-12,0
Confess	-8,-8

One can clearly see that Confess is still a better option for Player 2 since $-8 > -12$

Since the game is symmetric, similar is the outcome of an analysis made on the Player 2.

1.3.3 Battle of the Sexes

A husband and wife wish to go out together. They decided on Ballet and Football but forgot they had finalized which one. One person prefers Ballet and the other prefers Football. If they go to different places, each of them is equally unhappy not being able to be together.
The normal form of the game is:

	Ballet	Football
Ballet	2,1	0,0
Football	0,0	1,2

At this point it is very important to first develop the concept of a Nash equilibrium to continue our search for optimal strategy.

1.3.4 Nash Equilibrium

What actions will be chosen by the players in a strategic game? We wish to assume, as in the theory of a rational decision-maker, that each player chooses the best available action. In a game, the best action for any given player depends, in general, on the other players actions. So when choosing an action a player must have in mind the actions the other players will choose. That is, she must form a belief about the other players actions.

A **Nash equilibrium** is an action profile a^* with the property that no player i can do better by choosing an action different from a_i^* , given that every other player j adheres to a_j^* .

We can state the definition of a Nash equilibrium precisely:
Let a be an action profile, in which the action of each player i is a_i.
Let a_{-i} be any action of player i (either equal to a_i, or different from it). Then $(a_i^{'}$, $a_i)$ denotes the action profile in which every player j except i chooses her action a_j as specified by a, whereas player i chooses $a_i^{'}$. (The -i subscript on a stands for except i.) That is, $(a_i^{'}$, $a_{-i})$ is the action profile in which all the players other than i adhere to a while i deviates to $a_i^{'}$. (If $a_i^{'} = a_i$ then of course $(a_i^{'}$, $a_{-i}) = (a_i, a_{-i})$ = a). If there are three players, for example, then $(a_2^{'}, a_{-2})$ is the action profile in which players 1 and 3 adhere to a (player 1 chooses a_1, player 3 chooses a_3) and player 2 deviates to $a_2^{'}$.

Definition:

The action profile a^* in a strategic game is a Nash equilibrium if, for every player i and every action a_i of player i, a^* is at least as good according to player is preferences as the action profile $(a_i$, $a_i^*)$ in which player i chooses a_i while every other player j chooses a_j^* . Equivalently, for every player i,

$$u_i(a^*) \geq u_i(a_i, a_{-i}^*) \tag{1.1}$$

for every action a_i of player i, where u_i is a payoff function that represents player is preferences.

Examples

- In Prisoner's Dilemma, the Nash equilibrium is the action profile (Confess,Confess).
  In particular, the incentive to free ride (with 0 payoff, the maximal in the game)eliminates the possibility that the mutually desirable outcome (Keep Quiet,Keep Quiet) occurs.

- In Battle of the Sexes:
  Consider the action profiles (Ballet,Ballet) , (Ballet,Football) , (Football,Ballet) , (Football,Football).

  - (Ballet, Ballet): If player 1 switches to Football then her payoff decreases from 2 to 0; if player 2 switches to Football then her payoff decreases from 1 to 0. Thus a deviation by either player decreases her payoff. Thus (Bach, Bach) is a Nash equilibrium.
  - (Ballet, Football): If player 1 switches to Football then her payoff increases from 0 to 1. Thus (Ballet, Football) is not a Nash equilibrium. (Player 2 can increase her payoff by deviating, too, but to show the pair is not a Nash equilibrium it suffices to show that one player can increase her payoff by deviating.)
  - (Football, Ballet): If player 1 switches to Bach then her payoff increases from 0 to 2. Thus (Football, Ballet) is not a Nash equilibrium.
  - (Football, Football): If player 1 switches to Ballet then her payoff decreases from 1 to 0; if player 2 switches to Bach then her payoff decreases from 2 to 0. Thus a deviation by either player decreases her payoff. Thus (Football, Football) is a Nash equilibrium.

- In Matching Pennies, there is no Nash equilibrium.

Strict and Weak dominance

Strict dominance
In any game, a players action strictly dominates another action if it is superior, no matter what the other players do.
DEFINITION
In a strategic game , player is action $a_i^{''}$ strictly dominates her action $a_i^{'}$ if

$$u_i(a_i^{''}, a_{-i}) > u_i(a_i^{'}, a_{-i}) \tag{1.2}$$

for every list a_{-i} of the other players actions, where u_i is a payoff function that represents player is preferences.

Weak dominance
In any game, a players action weakly dominates another action if the first action is at least as good as the second action, no matter what the other players do, and is better than the second action for some actions of the other players.
DEFINITION
In a strategic game with ordinal preferences, player is action $a_i^{''}$ weakly dominates her action $a_i^{'}$ if

$$u_i(a_i^{''}, a_{-i}) \geq u_i(a_i^{'}, a_{-i}) \tag{1.3}$$

for every list a_{-i} of the other players actions and

$$u_i(a_i^{''}, a_{-i}) > u_i(a_i^{'}, a_{-i}) \tag{1.4}$$

for some list a_{-i} of the other players actions, where u_i is a payoff function that represents player is preferences.

1.4 Mixed Strategy Nash Equilibrium

Here is a very simple game. You and I each have a penny. Simultaneously, we choose whether to put our penny on the table with heads or tails facing up. If both of the pennies are heads or both of the pennies are tails (that is, they match), then you have to pay me a dollar. But if one is heads and the other is tails (that is, they do not match), then I have to pay you a dollar. The matrix form of the game is:

	Heads	Tails
Heads	1,-1	-1,1
Tails	-1,1	1,-1

Matching pennies is an example of a strictly competitive or zero sum game which has no Nash equilibrium.
Does that mean this game has no solution? Not nearly. As it turns out, every finite game has at least one Nash equilibrium. (John Forbes Nash, of A Beautiful Mind fame, proved this result, which is why we call this Nashs theorem. It is also why we search for **Nash** equilibria.) We define a game as being finite if the number of players is finite and the number of pure strategies each player has is finite. Here, there are two players, and each player has two pure strategies. Therefore, matching pennies is a finite game, so it must have a Nash equilibrium. We know matching pennies has no pure strategy Nash equilibria, so it must have an equilibrium in mixed strategies.

Suppose that player 2 chooses each of her actions with probability $\frac{1}{2}$

- If player 1 chooses Head with probability p and Tail with probability 1 - p then each outcome (Head, Head) and (Head, Tail) occurs with probability $\frac{1}{2}$ p, and each outcome (Tail, Head) and (Tail, Tail) occurs with probability $\frac{1}{2}$(1 - p). Thus player 1 gains 1 with probability $\frac{1}{2}$p + $\frac{1}{2}$(1 - p), which is equal to $\frac{1}{2}$, and loses 1 with probability $\frac{1}{2}$.
- In particular, the probability distribution over outcomes is independent of p! Thus every value of p is optimal. In particular, player 1 can do no better than choose Head with probability $\frac{1}{2}$ and Tail with probability $\frac{1}{2}$.
- A similar analysis shows that player 2 optimally chooses each action

with probability $\frac{1}{2}$ when player 1 does so. We conclude that the game has a stochastic steady state in which each player chooses each action with probability $\frac{1}{2}$.

We refer to a probability distribution over strategies as a mixed strategy and strategies with certainty as pure strategy.

1.4.1 Definition

The mixed strategy profile α^* in a strategic game is a (mixed strategy) Nash equilibrium if, for each player i and every mixed strategy α_i of player i, the expected payoff to player i of α^* is at least as large as the expected payoff to player i of $(\alpha_i\ ,\ \alpha^*_{-i})$ according to a payoff function whose expected value represents player is preferences over lotteries. Equivalently, for each player i,

$$U_i(\alpha^*) \geq U_i(\alpha_i, \alpha^*_{-i}) \tag{1.5}$$

for every mixed strategy α_i of player i, where $U_i(\alpha)$ is player is expected payoff to the mixed strategy profile α.

1.4.2 Expected Payoff

A players expected payoff to the mixed strategy profile α is a weighted average of her expected payoffs to all mixed strategy profiles of the type (a_i, α_{-i}), where the weight attached to (a_i, α_{-i}) is the probability $\alpha_i(a_i)$ assigned to a_i by player is mixed strategy α_i.

$$U_i(\alpha) = \sum_{a_i \in A_i} \alpha_i(a_i) U_i(a_i, \alpha_i), \tag{1.6}$$

where Ai is player is set of actions (pure strategies) and $U_i(a_i, \alpha_{-i})$ is her expected payoff when she uses the pure strategy that assigns probability 1 to a_i and every other player j uses her mixed strategy α_j.

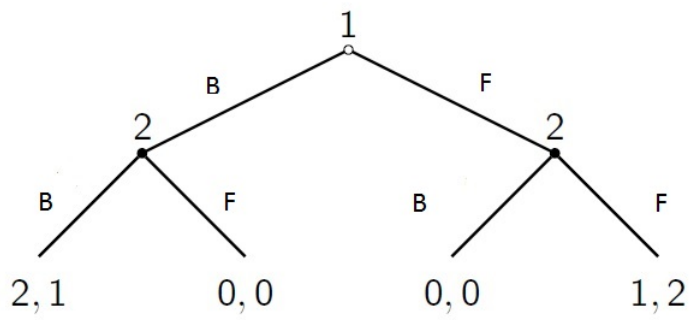
1.5 Bayesian Games: Games with incomplete information

An ASSUMPTION underlying the notion of Nash equilibrium is that each player holds the correct belief about the other players actions. To do so, a player must know the game she is playing; in particular, she must know the other players preferences. In many situations the participants are not perfectly informed about their opponents characteristics: bargainers may not know each others valuations of the object of negotiation, firms may not know each others cost functions, combatants may not know each others strengths, and jurors may not know their colleagues interpretations of the evidence in a trial. In some situations, a participant may be well informed about her opponents characteristics, butmay not know how well these opponents are informed about her own characteristics.

1.5.1 Introduction to incomplete information with a game

For sequential games, extensive form representation is considered to be the simplest and so here I show the extensive form of Battle of the Sexes, where B and F are Ballet and Football respectively:

Figure 1.2: Battle of the Sexes



- The battle of the sexes game, is a complete information "coordination" game.
- Both parties want to meet, but they have different preferences on "Ballet" and "Football".

	Ballet	Football
Ballet	2,1	0,0
Football	0,0	1,2

- Now imagine that player 1 does not know whether player 2 wishes to meet or wishes to avoid player 1. Therefore, this is a situation of **incomplete information** - also sometimes called asymmetric information.
- We represent this by thinking of player 2 having two different types, one type that wishes to meet player 1 and the other wishes to avoid him.
- More explicitly, suppose that these two types have probability 1/2 each. Then the game takes the form one of the following two with probability 1/2.

	Ballet	Football
Ballet	2,1	0,0
Football	0,0	1,2

And

	Ballet	Football
Ballet	2,0	0,2
Football	0,1	1,0

- From player 1's point of view, player 2 has two possible types (or equivalently, the world has two possible states each with 1/2 probability and only player 2 knows the actual state).
- We use Nash Equilibrium concept in an expanded game, where each different type of player 2 has a different strategy.
- Let us consider the following strategy profile (B; (B; F)), which means that player 1 will play B, and while in state 1, player 2 will also play B (when she wants to meet player 1) and in state 2, player 2 will play F (when she wants to avoid player 1).
- Clearly, given the play of B by player 1, the strategy of player 2 is a best response.
- Let us now check that player 2 is also playing a best response.
- Since both states are equally likely, the expected payoff of player 2 is

$$E[B; (B; F)] = \frac{1}{2} * 2 + \frac{1}{2} * 0 = 1 \tag{1.7}$$

- If, instead, he deviates and plays F, his expected payoff is

$$E[F;(B;F)] = \frac{1}{2} * 0 + \frac{1}{2} * 1 = \frac{1}{2} \tag{1.8}$$

- Therefore, the strategy profile (B; (B; F)) is a (Bayesian) Nash equilibrium.
- Interestingly, meeting at Football, which is the preferable outcome for player 2 is no longer a Nash equilibrium. Why not?
- Suppose that the two players will meet at Football when they want to meet. Then the relevant strategy profile is (F; (F;B)) and

$$E[F;(F;B)] = \frac{1}{2} * 1 + \frac{1}{2} * 0 = \frac{1}{2} \tag{1.9}$$

- If, instead, player 1 deviates and plays B, his expected payoff is

$$E[B;(F;B)] = \frac{1}{2} * 0 + \frac{1}{2} * 2 = 1 \tag{1.10}$$

- Therefore, the strategy profile (F; (F;B)) is not a (Bayesian) Nash equilibrium.

1.5.2 Definition

A Bayesian game consists of:

- A set of players I;
- A set of actions (pure strategies) for each player i : S_i ;
- A set of types for each player i : $\theta_i \in \Theta_i$;
- A payoff function for each player i : $u_i(s_1; ...; s_I; \theta_1; ...; \theta_I)$;
- A (joint) probability distribution p($\theta_1; ...; \theta_I$) over types (or P($\theta_1; ...; \theta_I$) when types are not finite).

Given p($\theta_1; ...; \theta_I$) we can compute the conditional distribution P($\theta_1; ...; \theta_I$) using Bayes rule.

- Hence the label "**Bayesian games**".
- Equivalently, when types are not finite, we can compute the conditional distribution P($\theta_{-i}|\theta_i$) given P($\theta_1; ...; \theta_I$).
- Since the payoff functions, possible types, and the prior probability distribution are common knowledge, we can compute expected payoffs of player i of type θ_i as:

$$U\left(s_i^{'}, s_{-i}, \theta_i\right) = \sum_{\theta_{-i}} p(\theta_{-i}|\theta_i) u_i(s_i^{'}, s_{-i}, \theta_i, \theta_{-i}) \tag{1.11}$$

when types are finite.

$$U\left(s_i^{'}, s_{-i}, \theta_i\right) = \int u_i(s_i^{'}, s_{-i}, \theta_i, \theta_{-i}) P(d\theta_{-i}|\theta_i) \tag{1.12}$$

when types are not finite.

Chapter 2

Quantum Games

2.1 Introduction

Eisert et al. have developed the method which allows, given some classical game, to construct its quantum counterpart. As an example they considered the 2-player 2-strategies classical game(Prisoner's Dilemma) and proposed the quantum version based on the geometry of SU(2) group.
But what does it mean for a game to be "**quantum**" ?

To define the same we begin with the so called :

2.1.1 Quantisation Rules for Quantum Games:

Introduce the following two notions in a classical game:

- Quantum Interference
- Quantum Entanglement

2.2 Quantum Penny Flip game

Meyer describes his game as follows. The starship Enterprise is facing some imminent catastrophe when Q appears on the bridge and offers to rescue the ship if Captain Picard[1] can beat him at a simple game: Q produces a penny and asks Picard to place it in a small box, head up. Then Q, followed by Picard, followed by Q, reaches into the box, without looking at the penny, and either flips it over or leaves it as it is. After Q's second turn they open the box and Q wins if the penny is head up. Q wins every time they play, using the following strategy:

$$
\begin{aligned}
&|0\rangle \xmapsto[\hat{H}]{\text{Q}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\
&\xmapsto[\hat{\sigma}_x \text{ or } \hat{I}_2]{\text{Picard}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\
&\xmapsto[\hat{H}]{\text{Q}} |0\rangle
\end{aligned}
$$

Here 0 denotes 'head' and 1 denotes 'tail', $\hat{H} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the Hadamard transformation , $\hat{I}_2$ means leaving the penny alone and the action with $\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ flips the penny over. Q's quantum strategy of putting the penny into the equal superposition of 'head' and 'tail', on his first turn, means that whether Picard flips the penny over or not, it remains in an equal superposition which Q can rotate back to 'head' by applying $\hat{H}$ again since $\hat{H} = \hat{H}^{-1}$. So Q always wins when they open the box.

Q's classical strategy consists of implementing $\hat{\sigma}_x$ or $\hat{I}_2$ on his turn. When Q is restricted to play only classically, flipping the penny over or not on each turn with equal probability becomes an optimal strategy for both the players. By adapting this classical strategy Q wins only with probability $\frac{1}{2}$. By using a quantum strategy Q can, therefore, win with probability 1.

[1] Meyer considered the initials and abilities of Picard and Q ideal for his illustration.

2.3 Quantum Prisoner's Dilemma

Beginning with the already discussed classical game Prisoner's dilemma, I demonstrate that

1. There exists a particular pair of quantum strategies which always gives reward and is a Nash equilibrium and
2. There exist a particular quantum strategy which always gives at least reward if played against any classical strategy.

The quantum formulation proceeds by assigning the possible outcomes of the classical strategies D and C two basis vectors $|D\rangle$ and $|C\rangle$ in the Hilbert space of a two state system, i.e., a qubit. At each instance, the state of the game is described by a vector in the tensor product space which is spanned by the classical game basis $|CC\rangle$, $|CD\rangle$, $|DC\rangle$ and $|DD\rangle$, where the first and second entry refer to Alices and Bobs qubit, respectively.

We denote the games initial state by $|\psi_0\rangle = \hat{J}\ |CC\rangle$, where J is a unitary operator which is known to both players. For fair games $\hat{J}$ must be symmetric with respect to the interchange of the two players. The strategies are executed on the distributed pair of qubits in the state $|\psi_0\rangle$. Strategic moves of Alice and Bob are associated with unitary operators U_A and U_B, respectively, which are chosen from a strategic space S. The independence of the players dictates that U_A and U_B operate exclusively on the qubits in Alices and Bobs possession, respectively. The strategic space S may therefore be identified with some subset of the group of unitary 2 X 2 matrices SU(2).
Having executed their moves, which leaves the game in a state $U_A \otimes U_B \hat{J}|CC\rangle$, then we disentangle the states using $\hat{J}^\dagger$. Alice and Bob forward their qubits for the final measurement which determines their payoff.
So the final state of the qubits after the game is:

$$|\psi_f\rangle = \hat{J}^\dagger U_A \otimes U_B \hat{J}|CC\rangle \tag{2.1}$$

Yet quantum mechanics being a fundamentally probabilistic theory, the only strategic notion of a pay-off is the expected pay-off. Alices expected pay-off is given by:

$$\$_A = rP_{CC} + pP_{DD} + tP_{DC} + sPCD \tag{2.2}$$

where p,r,s,t are the classical payoffs from the payoff matrix.
where $P_{\sigma\sigma'} = \langle\sigma\sigma'|\psi_f\rangle$is the joint probability that the channels $\sigma$and σ' of

the Stern-Gerlach type devices will click.

Note that Alices expected pay-off $\$_A$ not only depends on her choice of strategy U_A, but also on Bobs choice U_B. It proves to be sufficient to restrict the strategic space to the 2-parameter set of unitary 2 x 2 matrices.

$$U(\theta,\phi) = \begin{pmatrix} e^{i\phi}\cos\frac{\theta}{2} & \sin\frac{\theta}{2} \\ -\sin\frac{\theta}{2} & e^{-i\phi}\cos\frac{\theta}{2} \end{pmatrix} \tag{2.3}$$

with $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq \frac{\pi}{2}$

To be specific, we associate the strategy cooperate with the operator
C = U(0,0); C = $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
while the strategy defect is associated with a spin-flip,
D = U(π,0); D = $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ In order to guarantee that the ordinary Prisoners Dilemma is faithfully represented, we impose the subsidiary conditions.

In order to guarantee that the ordinary Prisoners' Dilemma is faithfully represented, we impose the subsidiary conditions

$$[\hat{\mathcal{J}}, \hat{D}\otimes\hat{D}] = 0\,, [\hat{\mathcal{J}}, \hat{D}\otimes\hat{C}] = 0\,, [\hat{\mathcal{J}}, \hat{C}\otimes\hat{D}] = 0\,. \tag{2.4}$$

These conditions together with the identification $\tilde{\mathcal{J}} = \hat{\mathcal{J}}^\dagger$ imply that for any pair of strategies taken from the subset $S_0 \equiv \{\hat{U}(\theta,0)|\,\theta\in[0,\pi]\}$, the joint probabilities $P_{\sigma\sigma'}$ factorize, $P_{\sigma\sigma'} = p_A^{(\sigma)} p_B^{(\sigma')}$, where $p^{(C)} = \cos^2(\theta/2)$ and $p^{(D)} = 1 - p^{(C)}$. Identifying $p^{(C)}$ with the individual preference to cooperate, we observe that condition in fact assures that the *quantum* Prisoners' Dilemma entails a faithful representation of the most general *classical* Prisoners' Dilemma, where each player uses a biased coin in order to decide whether he or she chooses to cooperate or to defect . Of course, the entire set of quantum strategies is much bigger than S_0, and it is the quantum sector $S\backslash S_0$ which offers additional degrees of freedom that can be exploited for strategic purposes. Note that our quantization scheme applies to any two player binary choice symmetric game and – due to the classical correspondence principle is to a great extent canonical.

2.4 Parrondo's paradox

Inspired by the flashing Brownian ratchet, Parrondos games present an apparently paradoxical situation. The games can be realized as coin tossing events. Game A uses a single biased coin while game B uses two biased coins and has a state dependent rule based on the players current capital. Playing each of the games individually causes the player to lose. However, a winning expectation is produced when randomly mixing games A and B. This phenomenon is investigated and mathematically analyzed to give explanations on how such a process is possible.

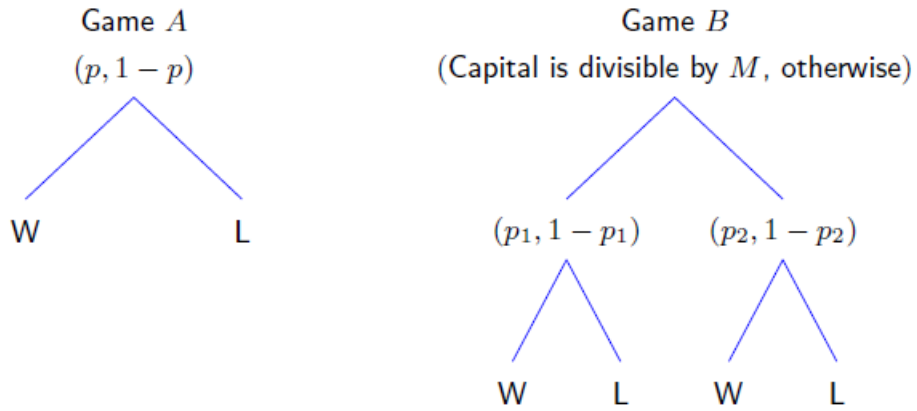
2.4.1 Introduction

With Parrondos games, which can be also played using simple biased coins, we go one step further to generate a positively biased outcome from a combination of two negatively biased processes. This type of behavior is not impossible and exists in other fields. In control theory, the combination of two unstable systems can cause them to become stable.
The inspiration of Parrondos games came from the Spanish physicist Juan M. R. Parrondo. He used the games as a pedagogical illustration of the Brownian ratchet. In one of his fields of interest, Brownian motors, he was able to see a link between mathematical games of chance and the Brownian ratchet. Using the equations for detailed balance, analogous to chemical reaction rates, enabled Parrondo to devise some working probabilities.

2.4.2 Construction of the games

Game A is straight forward and consists of a single branch element. The probability of winning and losing is p and 1 - p respectively. Game B is a little more complex, as a game rule is first required to chose which of two coins to toss. It is described by the following statement. If the present capital is a multiple of an integer M, then the chance of winning is p_1, if the capital is not a multiple of M, then the chance of winning is p_2. The losing probabilities are 1-p_1 and 1-p_2, respectively. Thus, game B uses two coins, the rule of which one to use depends on the value of the current capital. For future reference let the coins B_1 and B_2 have winning probabilities p_1 and p_2 respectively. If we require control of the three probabilities p, p_1 and p_2 via a single variable, a biasing parameter ϵ can be used to represent a subset of the parameter space with the transformation

Figure 2.1: Diagrammatic representation of the game with W for winning and L for losing



p = $\frac{1}{2}$ - ϵ,
$p_1 = \frac{1}{10}$ - ϵ and
$p_2 = \frac{3}{4}$ - ϵ
This parameterization along with M = 3 gives Parrondos original numbers for the games that he constructed.
It can be deduced via a detailed balance calculation and simulations, that both game A and game B lose when is greater than zero. Consider the scenario if we start switching between the two losing games, play two games of A, two games of B, two of A, and so on for example. The result, which is quite counter intuitive, is that we start winning! That is, we can play the two losing games A and B in such a way as to produce a winning expectation. Furthermore, deciding which game to play next by tossing a fair coin also yields a winning expectation.

2.4.3 Brownian ratchet

The aim is to convert the physical system of the Brownian ratchet to a mathematical system. One method is to use simple gambling games. The quantities of the Brownian ratchet must be mapped to analogous mathematical variables. The capital can be used in lieu of particles in the ratchet and the probabilities in lieu of energy potentials. Initially this will be adequate to transform the system.

The easy job is to convert the flat potential, which allows diffusion equally

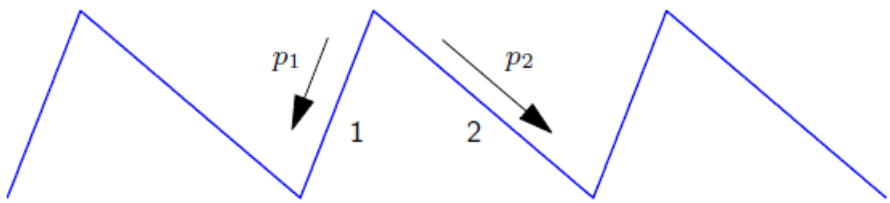


Figure 2.2: Diagrammatic representation of the game with p1 and p2 represented as slopes

in both directions. This is simply a fair game using a single probability, p say. The sawtooth potential appears tricky at first, but on closer inspection is not too difficult. The sawtooth shape shown in Fig. 3 can be considered piece-wise linear, consisting of a steep positive slope (segment 1) and a gentle negative slope (segment 2). This suggests that in order to replicate this mathematically two games are required, one for each of the two segments, remembering that this represents a potential and particles will fall downwards. Segment 1 requires a game with a strong losing probability ($1p_1$, say) and segment 2 requires a game with a weak winning probability (p_2, say). In the Brownian ratchet the particles know what segment to follow due to their spatial location, i.e. there is spatial dependency. This attribute can be mapped to the games by use of an if statement. If the capital is within a certain range play the first coin (coin 1), otherwise play the second coin (coin 2). This needs to be periodic, which invites the use of a modulo operator. The capital is an integral amount so the modulo can be formed over a range of capital, say M. Then within a certain range play coin 1, or if in the remaining part of the range play coin 2.

2.4.4 Explanation

An alternative explanation of the two systems can be given in terms of the localization of particles or capital at system ceilings. Considering game B, the capital tends to localize between the Mn-1 and Mn states for an integer n that represents the subsystems. This is due to the chosen probabilities of p_1 and p_2. At Mn-1, there is a high probability (p_2) the capital will increase to Mn and at that state there is an even higher probability (1-p_1) the capital will be pushed back down to Mn-1. Due to the oscillatory behavior between these two states and the tendency for capital to drift upwards when in states (M(n-1)+1) to (Mn-1), there is a localization of capital at the Mn ceilings. In the same way, the particles in the ratchet teeth are localized to the pits, a small movement is met with an opposing long gentle or short steep edge. Thus, in each subsystem an extra kick is required to move the capital or particles backwards or forwards one period.
Adding game A to the playing sequence improves the chance of moving up to the next subsystem because most of the capital is localized at these ceilings. Switching to an approximately fair game allows almost half of the capital at these ceilings to move up to the next subsystem (i.e. from Mn to M(n+1)), while the other half moves down a few steps only to be brought back to the Mn ceiling when game B is played again. This is exactly what happens when the ratchet teeth are made to disappear in the flashing Brownian ratchet about half of the particles can easily move over the steep edge into the next pit while the remaining particles fall back into the same pit via the gentle edge when the ratchet teeth appear again.

2.5 Quantum Parrondo's Games

A physical interpretation of our system could be a collection of polarised photons where $|0\rangle$ represents horizontal polarisation and $|1\rangle$ represents vertical polarisation (though we could just as easily consider instead the spin of a spin one-half particle).

In classical gambling games there is a random element, and in a Parrondo's game the results of the random process is used to alter the evolution of the game. The quantum mechanical model is deterministic until a measurement is made at the end of the process. The element of chance that is necessary in the classical game is replaced by a superposition that represents all the possible results in parallel. We can get new behaviour by the addition of phase factors in our operators and by interference between states. A further random element can be introduced, in future studies, by perturbing the system with noise or by considering decoherence during the evolution of the sequence of games.

An arbitrary $SU(2)$ operation on a qubit can be written as

$$\begin{aligned} \hat{A}(\theta,\gamma,\delta) &= \hat{P}(\gamma)\,\hat{R}(\theta)\,\hat{P}(\delta) \\ &= \begin{pmatrix} e^{-i(\gamma+\delta)/2}\cos\theta & -e^{-i(\gamma-\delta)/2}\sin\theta \\ e^{i(\gamma-\delta)/2}\sin\theta & e^{i(\gamma+\delta)/2}\cos\theta \end{pmatrix} , \end{aligned} \tag{2.5}$$

where $\theta \in [-\pi,\pi]$ and $\gamma,\delta \in [0,2\pi]$. This is our game A: the quantum analogue of a single toss of a biased coin. One way of achieving this physically on a polarised photon would be to sandwich a rotation of the plane of polarisation by θ (R) between two birefringent media (P) that introduce phase differences of γ and δ, respectively, between the horizontal and vertical planes of polarisation. Game B consists of four $SU(2)$ operations, each of the form of , whose use is controlled by the results of the previous two games:

$$\begin{aligned} &\hat{B}(\phi_1,\alpha_1,\beta_1,\phi_2,\alpha_2,\beta_2,\phi_3,\alpha_3,\beta_3,\phi_4,\alpha_4,\beta_4) = \\ &\begin{bmatrix} A(\phi_1,\alpha_1,\beta_1) & 0 & 0 & 0 \\ 0 & A(\phi_2,\alpha_2,\beta_2) & 0 & 0 \\ 0 & 0 & A(\phi_3,\alpha_3,\beta_3) & 0 \\ 0 & 0 & 0 & A(\phi_4,\alpha_4,\beta_4) \end{bmatrix} . \end{aligned} \tag{2.6}$$

This acts on the state

$$|\psi(t-2)\rangle \otimes |\psi(t-1)\rangle \otimes |i\rangle \ , \tag{2.7}$$

where $|\psi(t-1)\rangle$ and $|\psi(t-2)\rangle$ represent the results of the two previous games and $|i\rangle$ is the initial state of the target qubit. That is,

$$\hat{B}|q_1q_2q_3\rangle = |q_1q_2b\rangle \ , \tag{2.8}$$

where $q_1, q_2, q_3 \in \{0, 1\}$ and b is the output of the game B.

The results of n successive games of B can be computed by

$$\begin{aligned} |\psi_f\rangle &= (\hat{I}^{\otimes n-1} \otimes \hat{B})(\hat{I}^{\otimes n-2} \otimes \hat{B} \otimes \hat{I})(\hat{I}^{\otimes n-3} \otimes \hat{B} \otimes \hat{I}^{\otimes 2}) \\ &\quad \ldots (\hat{I} \otimes \hat{B} \otimes \hat{I}^{\otimes n-2})(\hat{B} \otimes \hat{I}^{\otimes n-1}) \, |\psi_i\rangle \ , \end{aligned} \tag{2.9}$$

with $|\psi_i\rangle$ being an initial state of $n+2$ qubits. The first two qubits of $|\psi_i\rangle$ are left unchanged and are only necessary as an input to the first game of B. In this , $\hat{I}$ is the identity operator for a single qubit. The result of other game sequences can be computed in a similar manner. The simplist case to study is that of two games of A followed by one game of B, since the results of one set of games do not feed into the next. The sequence AAB played n times results in the state

$$\begin{aligned} |\psi_f\rangle &= \left(\hat{I}^{\otimes 3n-3} \otimes (\hat{B}(\hat{A} \otimes \hat{A} \otimes \hat{I}))\right) \\ &\qquad \left(\hat{I}^{\otimes 3n-6} \otimes (\hat{B}(\hat{A} \otimes \hat{A} \otimes \hat{I})) \otimes \hat{I}^{\otimes 3}\right) \\ &\qquad \ldots \left((\hat{B}(\hat{A} \otimes \hat{A} \otimes \hat{I})) \otimes \hat{I}^{\otimes 3n-3}\right) \, |\psi_i\rangle \\ &= \hat{G}^{\otimes n} |\psi_i\rangle \ , \end{aligned} \tag{2.10}$$

where $\hat{G} = \hat{B}(\hat{A} \otimes \hat{A} \otimes \hat{I})$ and $|\psi_i\rangle$ is an initial state of $3n$ qubits.

In quantum game theory the standard protocol is to take the initial state $|00\ldots0\rangle$, apply an entangling gate, then the operators associated with the players strategies and finally a dis-entangling gate. A measurement on the resulting state is taken and then the payoff is determined. If the entangling gate depends upon some parameter, then the classical game can be reproduced when this parameter is set to zero, representing no entanglement. In the present case this is problematic since the entangling gate $\hat{J}$ used by Eisert and others does not commute with the classical limit (all phases $\to 0$) of $\hat{B}$, which was Eisert's motivation for the choice of $\hat{J}$. Thus this protocol would not reproduce the classical game when the phases are set to zero. So instead we follow and suppose the initial state is already in the maximally entangled state:

$$|\psi_i^m\rangle = \frac{1}{\sqrt{2}}(\, |00\ldots0\rangle \, + \, |11\ldots1\rangle \,) \ . \tag{2.11}$$

The classical game can be reproduced by choosing the alternative initial state $|\psi_i\rangle = |00\ldots0\rangle$. Thus the classical game is still a subset of the quantum one. If $|\psi_i\rangle$ is a superposition, interference effects that either enhance or reduce the success of the player can be obtained. The addition of non-zero phases in the operators $\hat{A}$ and $\hat{B}$ can modify this interference.

To determine the payoff let the payoff for a $|1\rangle$ state be one, and for a $|0\rangle$ state be negative one. The expectation value of the payoff from a sequence of games resulting in the state $|\psi_f\rangle$ can be computed by

$$\langle \$ \rangle = \sum_{j=0}^{n} \left((2j-n) \sum_{j'} \left| \langle \psi_j^{j'} | \psi_f \rangle \right|^2 \right) , \tag{2.12}$$

where the second summation is taken over all basis states $\langle \psi_j^{j'} |$ with j 1's and $n-j$ 0's.

2.5.1 Results

Consider the game sequence AAB. With an initial state of $|000\rangle$ this yields a payoff of

$$\begin{aligned} \langle \$^0_{AAB} \rangle &= \sin^4\theta\,(2-\cos 2\phi_4) \;-\; \cos^4\theta\,(2+\cos 2\phi_1) \\ &\quad - \frac{1}{4}\sin^2 2\theta\,(\cos 2\phi_2 + \cos 2\phi_3) \;, \end{aligned} \tag{2.13}$$

which is the same as the classical result. In order to get interference there needs to be two different ways of arriving at the same state. We need only choose some superposition not the maximally entangled state, however this is the most interesting initial state to study. Choosing $|\psi_i^m\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ the result is

$$\begin{aligned} \langle \$^m_{AAB} \rangle &= \frac{1}{2}\cos 2\theta\,(\cos 2\phi_4 - \cos 2\phi_1) \\ &\quad + \frac{1}{4}\sin^2 2\theta\;(\cos(2\delta+\beta_1)\,\sin 2\phi_1 \\ &\quad - \cos(2\delta+\beta_2)\,\sin 2\phi_2 \;-\; \cos(2\delta+\beta_3)\,\sin 2\phi_3 \\ &\quad + \cos(2\delta+\beta_4)\,\sin 2\phi_4) \;. \end{aligned} \tag{2.14}$$

It is the dependence on the phase angles δ and β_i that can produce a result that cannot be obtained in the classical game. In the quantum case a range of payoffs can be obtained for a given set of θ and ϕ_i's, that is, for a given set of probabilities for games A and B.

The probabilities given yield a situation where both games A and B are individually losing but the combination of A and B can produce a net positive payoff provided $\epsilon < 1/168$. With the quantum version of the games the expectation value of the payoff (to $O[\epsilon]$) for a single sequence of AAB can vary between $0.812 + 0.24\epsilon$ and $-0.812 + 0.03\epsilon$. The maximum result is obtained by setting $\beta_2 = \beta_3 = \pi - 2\delta$ and $\beta_1 = \beta_4 = -2\delta$, while the

minimum is obtained by $\beta_1 = \beta_4 = \pi - 2\delta$ and $\beta_2 = \beta_3 = -2\delta$. The values of the α_i's are not relevant. Classically AAB is a winning sequence provided $\epsilon < 1/112$.

The average payoff for the classical game sequence AAB_1 (that is, AAB where each branch of B is the best branch B_1) is $4/5 - 6\epsilon$ which is less than the greatest value of $\langle \$^m_{AAB} \rangle$. Thus the entanglement and the resulting interference can make game B in the sequence AAB better than its best branch taken alone. Indeed the expectation value for the payoff of a quantum AAB_1 on the maximally entangled initial state vanishes due to destructive interference. by setting all the ϕ_i's equal to ϕ_1 and all the β_i's to β_1.)

The quantum enhancement disappears when we play a sequence of AAB's on the maximally entangled initial state. In this case the phase dependent terms undergo destructive interference and we are left with a gain per qubit of order ϵ.

A sequence of B's leaves the first two qubits unaltered while a sequence of AB's leaves the first qubit unaffected. In these cases the final states that arise from $|\psi_i\rangle = |000\rangle$ and $|\psi_i\rangle = |111\rangle$ are distinct so a superposition of these two states produces no interference. An initial state that is a different superposition may give interference effects.

Chapter 3

Nonlocal Games

3.1 Vaidman's Game

I present the following game for a team of three players. The players are allowed to make any preparations before they are taken to three remote locations A,B and C. Then, at a certain time t, each player is asked one of two possible questions: What is X? or What is Y ? (see Fig. 1). Each player must give an answer which is also limited to only two possibilities: 1 or 1. They have to give their answer quickly, i.e., before the time they might receive a message sent by another player after the time t. According to the rules of the game, either all players are asked the X question, or only one player is asked the X question and the other two are asked the Y question. The team wins if the product of their three answers is 1 in the case of three X questions and is 1 in the case of one X and two Y questions. My friend is asked what should the team do in order to win for sure.

3.1.1 Analysing the game

Since each player is not able to get any message from the other players about which questions they were asked before the time she has to give an answer, it seems that she cannot gain anything by delaying the decision of which answer to give for each question until the question is actually asked.
Thus, an optimal strategy should correspond to prior definite decisions of each player which answers to give for possible questions. But it is easy to

prove that any such strategy cannot ensure winning for all allowed combinations of questions.
Indeed, if it does, then there must be a set of answers $X_A, Y_A, X_B, Y_B, X_C, Y_C$, where X_A is the answer of the player in A on question X, etc., such that the following equations are fulfilled.

$$X_A X_B X_C = -1, \tag{3.1}$$

$$X_A Y_B Y_C = 1, \tag{3.2}$$

$$Y_A X_B Y_C = 1, \tag{3.3}$$

$$Y_A Y_B X_C = 1. \tag{3.4}$$

This however is impossible, because the product of all left hand sides of equations is the product of squares of numbers which are +1 or -1 and therefore it equals 1, while the product of all right hand sides of these equation yields 1.

The particles are prepared in a entangled state (which is usually called the GHZ state):

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|1_z\rangle_A|1_z\rangle_B|1_z\rangle_C - |0_z\rangle_A|0_z\rangle_B|0_z\rangle_C). \tag{3.5}$$

Now, if a member of the team is asked the X question, she measures σ_x and gives the answer which she obtains in this experiment. If she is asked the Y question, she measures σ_y instead. Quantum theory ensures that the team following this strategy always wins.

The states satisfy the above games of the rule:

$$\sigma_{A_x}\sigma_{B_x}\sigma_{C_x} = -1 \tag{3.6}$$

$$\sigma_{A_x}\sigma_{B_y}\sigma_{C_y} = 1 \tag{3.7}$$

$$\sigma_{A_y}\sigma_{B_x}\sigma_{C_y} = 1 \tag{3.8}$$

$$\sigma_{A_y}\sigma_{B_y}\sigma_{C_x} = 1 \tag{3.9}$$

Let us show, for example, that the first equation is true. In the spin x bases for all particles the GHZ state (which is given in the spin z bases) is:

$$|GHZ\rangle = \tfrac{1}{4}[(|1_x\rangle_A + |0_x\rangle_A)(|1_x\rangle_B + |0_x\rangle_B)(|1_x\rangle_C + |0_x\rangle_C) - (|1_x\rangle_A - |0_x\rangle_A)(|1_x\rangle_B - |0_x\rangle_B)(|1_x\rangle_C - |0_x\rangle_C)]$$

which becomes

$$|GHZ\rangle = \frac{1}{2}[|1_x\rangle_A|1_x\rangle_B|0_x\rangle_C + |1_x\rangle_A|0_x\rangle_B|1_x\rangle_C) + |0_x\rangle_A|1_x\rangle_B|1_x\rangle_C + |1_x\rangle_A|0_x\rangle_B|0_x\rangle_C]$$

Therefore, we see explicitly that the GHZ state is a superposition of states for each of which $\sigma_{A_x}\sigma_{B_x}\sigma_{C_x} = -1$.

In the quantum solution of the problem the players do not decide in advance the answers theyll give for each out of the two possible questions. In the proof of the impossibility of this task presented above it was erroneously assumed that delaying the decision which answer to give until the time the question is asked cannot help. The assumption sounds plausible since relativistic causality prevented sending signals after the time the questions were asked, but, nevertheless, the assumption is wrong because it does not take into account unusual correlations which quantum objects can exhibit.

3.1.2 Why Vaidman's game is popular ?

Performing the game with spin$\frac{1}{2}$ particles as described above might serve as an experimental proof of the nonexistence of local hidden variables which produce agreement with experiment. Suppose, however, that Nature is governed by local hidden variables. This means that the results σ_{A_x}, σ_{A_y}, σ_{B_x}, σ_{B_y}, σ_{C_x}, σ_{C_y} exist prior to their measurements and, therefore, there are answers $X_A, Y_A, X_B, Y_B, X_C, Y_C$ prior to the time the questions are asked. These answers can fulfill at most three out of four equations (1) and, therefore, in the game in which the four question patterns are chosen with equal probability, on average there will be at most a 75 per cent success rate of any team playing this game. If a team shows on average a higher result, this is an experimental proof that Nature is not governed by local hidden variables.

3.2 Quantum games with a diffraction set-up

Historically speaking, the De Broglie's original idea that travelling material particles have waves associated with them was undoubtedly the key concept behind the development of quantum physics in early part of the twentieth century. Soon afterwards, Davisson and Germer verified the idea in their experimental demonstration of the diffraction of electrons by crystals. De Broglie's proposed that a travelling electron with momentum p has an associated wave of wavelength $\lambda = h/p$, where h is the Plank's constant. To make λ a measurable quantity, under normal laboratory conditions, the momentum p should have similar order of magnitude as h. h being a very small quantity shows why it is very hard to detect waves associated with macroscopic objects. Our motivation is to take this quantum feature – associating wave-like properties to micro objects – as a resource that can be used to play a quantum game. Such a quantum game can be realized using an apparatus consisting of travelling electrons, multiple slits intercepting them, and a resulting diffraction pattern. In this set-up a player's choice of a 'pure strategy' consists of opening or closing slits at his/her disposal. Suppose the apparatus is adjusted such that when λ approaches zero the classical game is reproduced. It can then be argued that because an observation of a value of λ quite away from zero is entirely a quantum feature, therefore, the resulting different payoffs for the players correspond to a quantum form of the classical game. In this setup the players' payoffs are to be found from the diffraction pattern formed on the screen. We show the possibility of finding a value for λ that makes appear a non-classical equilibrium in the PD when the players play only the pure strategies. The classical game remains a subset of its quantum version because with λ approaching zero the classical game is reproduced.

The motivation to play a quantum form of PD, without using the phenomenon of entanglement, also derives from Feynman's excellent exposition of quantum behavior of atomic objects. He describes and compares the diffraction patterns in two similar set-ups that are imaginary but experimentally realizable. The two set-ups consist of bullets and electrons passing through a wall with two slits. Feynman then describes the well-known quantum property – associating waves to all material particles – to distinguish the diffraction patterns of bullets and electrons. The disappearance of a pattern for the bullets, he explains, is due to tiny wavelengths of the associated waves. For such waves the pattern becomes very fine and, with a detector of finite size, one cannot distinguish the separate maxima and minima. We ask why not to play a game, in the Feynman's imaginary experimental set-up, such that the classical game corresponds when, in Feynman's words, bullets are fired; and a quantum game corresponds when electrons replace

the bullets.

3.2.1 Playing Prisoners' Dilemma

We select the PD to be played in a diffraction set-up. The classical PD, in its general form, is represented by the following matrix:

$$\begin{array}{cc} & \text{Bob} \\ & \begin{array}{cc} C & D \end{array} \\ \text{Alice} \begin{array}{c} C \\ D \end{array} & \begin{pmatrix} (r,r) & (s,t) \\ (t,s) & (u,u) \end{pmatrix} \end{array} \tag{3.10}$$

where $t > r > u > s$. To make the classical game imbedded in its quantum version the positive coefficients u, r, s and t, appearing in the matrix (3.10), are translated into distances between the slits. Each player is in control of two slits such that his/her strategy consists of opening one of the slits and closing the other. For example, if Alice decides to cooperate then she opens the slit C and closes the slit D. Because Bob has a similar choice, therefore, all possible moves by the players leads to the opening of two slits and closure of the other two, with the separation between the two open slits depending on the moves of the players. It happens when only the so-called 'pure-strategies' can be played by the players, which in the present setup means to open a slit and close the other. Now, at the final stage of the game, the action of the arbiter – responsible for finding the payoffs when the players have made their moves – consists of measuring the distance between two peaks of the diffraction pattern. This peak-to-peak distance is known to be λ/d, where d is the separation between the open sits and λ is the wavelength associated with the bombarded material objects, like electrons. Payoffs to the players are functions of λ/d and it, then, explains why it is useful to translate the coefficients of the matrix of the classical game into the separations d between the slits. When bullets are fired, which means the particles become heavier and corresponding λ is very nearly zero, the payoffs become classical and depend only on d i.e. the separation between the slits. A payoff representation in terms of λ/d contains both the classical and quantum aspects of the matrix game played in this set-up. The experimental set-up shown in the Fig. (3.1) sketches the diffraction set-up to play a quantum game.

For PD the payoffs are symmetric for the players and a single equation can describe the payoffs to both the players when their strategies are known. A usual way to express it is to write $P(s_1, s_2)$ for the payoff to the s_1-player

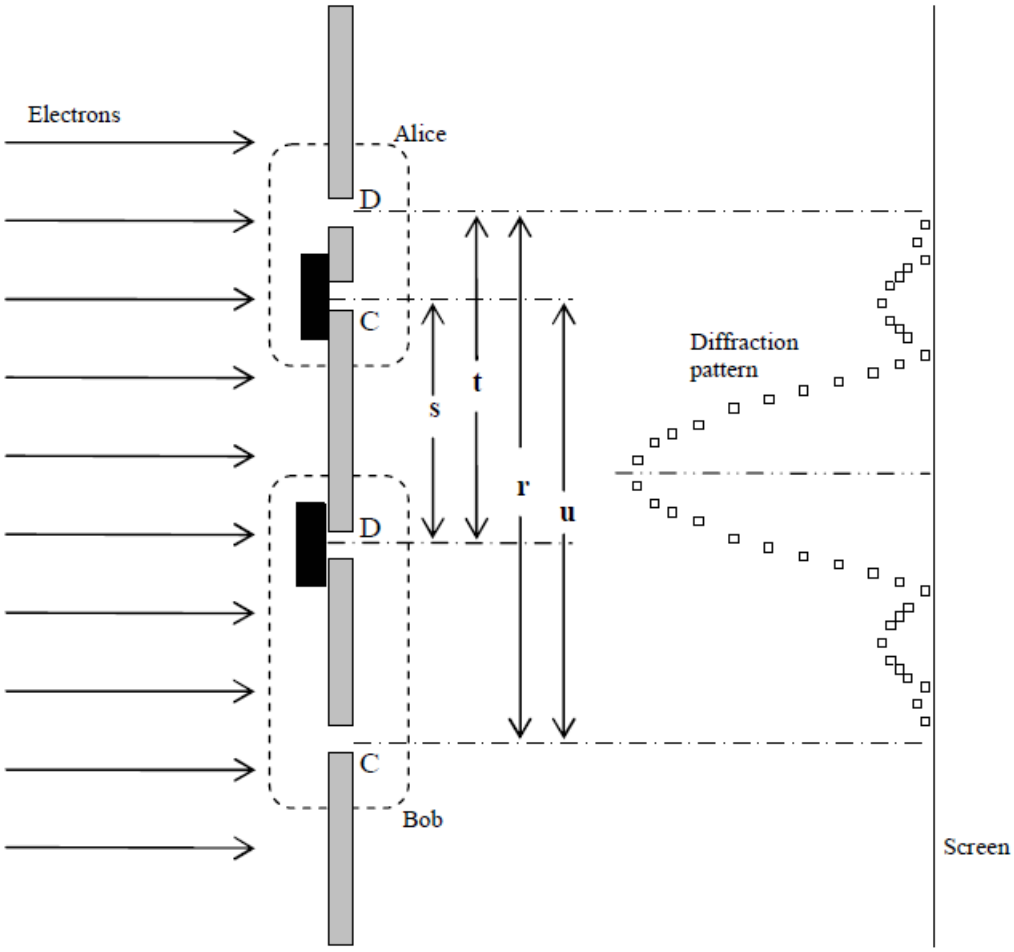


Figure 3.1: A multi-slit diffraction set-up to play a quantum form of Prisoners' Dilemma. A window with four slits faces an electron source. Each player has access to two slits. A player plays a pure strategy by opening a slit and closing the other. Referee finds the players' payoffs by measuring the peak-to-peak distance on the diffraction pattern formed on the screen.

against the s_2-player. Such a single equation representation is usually used in evolutionary games consisting of symmetric bi-matrix conflicts. The s_1-player is referred to as the 'focal' player and the s_2-player as just the 'other' player. The PD is one such example for which a single payoff equation can capture the essence of the idea of a symmetric NE. A strategy $s^\star$ is a symmetric NE if

$$P(s^\star, s^\star) - P(s, s^\star) \geq 0, \quad \text{for all } s \tag{3.11}$$

saying that the focal player cannot be better off by diverging away from $s^\star$. Because the set-up of the Fig. (3.1) involves coefficients in the classical payoff matrix corresponding to the first player, therefore, finding a symmetric NE becomes immediately possible when the first player is taken as focal. It also shows why writing payoff as $P(s_1, s_2)$ is relevant to the set-up of the Fig. (3.1). For example, classically the strategy of defection D is a symmetric NE because $P(D, D) - P(C, D) = (u - s) > 0$, where the players' moves consist of only the pure strategies.

In the set-up of Fig. (3.1) for every pure strategy move the players have option to play, a unique separation d between the slits is obtained that can have four possible values i.e. u, r, s or t. Classically $P(C, C) = r$, $P(C, D) = s$, $P(D, C) = t$ and $P(D, D) = u$. It is observed in the Fig. (3.1) that the classical payoff to the focal player, against the other, can be equated to the separation between the two open slits d.

Now assume that instead of simply $P(s_1, s_2) = d$ the arbiter uses the payoff equation:

$$P(s_1, s_2) = d + k(\lambda/d) \tag{3.12}$$

where k is a positive constant that can be called a *scaling factor*. $P(s_1, s_2)$, obviously, reduces to its classical counterpart when λ is very nearly zero. Suppose the strategy of cooperation C is a symmetric NE:

$$P(C, C) - P(D, C) = \{k\lambda(1/r - 1/t) - (t - r)\} \geq 0 \tag{3.13}$$

It requires $\lambda \geq rt/k$. For electrons of mass m travelling with velocity v it gives $v \leq (kh/mrt)$. Supposing r and t are both non-zero, the arbiter's problem consists of finding an appropriate value for the scaling factor k that brings v into a reasonable range from experimental point of view. When the electrons have associated wavelength $\lambda \geq rt/k$ the strategy of cooperation becomes a symmetric NE, and each player gets a payoff $r + k\lambda/r$. Similarly

when the pure strategy of defection is a symmetric NE in the quantum game, we have

$$P(D,D) - P(C,D) = \{-k\lambda(1/s - 1/u) + (u - s)\} \geq 0 \tag{3.14}$$

It requires $\lambda \leq su/k$. After the scaling factor k is decided the wavelength λ determines which pure strategy should be a symmetric NE. Two ranges for λ can be indicated as $\lambda \leq su/k$ and $\lambda \geq rt/k$. Defection and cooperation are symmetric NE for these ranges, respectively. Because the constants t, r, u, s and k are all positive the classical game is in the first range of λ. Non-classical equilibrium of cooperation shows itself in the second range of λ.

Du et al.'s recent analysis of the quantum PD, with players' access to Eisert's two-parameter set of unitary operators, has shown an intriguing structure in the game as a function of the amount of entanglement. The game becomes classical when the entanglement vanishes.

In the set-up of Fig. (3.1) the quantity λ behaves in a way similar to the amount of entanglement in Du et al.'s analysis . But this set-up is devoid of the notion of entanglement and relies instead on a different quantum aspect. An aspect which is as much 'quantum' in nature as the phenomenon of entanglement for qubit systems.

There is however a difference, to be noticed, between the set-ups of Eisert et al. and that of the Fig. (3.1). Players' actions in Eisert et al.'s set-up are quantum mechanical in nature in that they make moves with quantum operators. In the present set-up, on the contrary, the players' actions are entirely classical consisting of opening or closing slits. In a sense it is similar to the players' actions in Marinatto and Weber's scheme. In their scheme players possess quantum operators but they apply them on an initial quantum state with classical probabilities; so that the players' moves can be considered classical as well. It can be said that, apart from the work of Eisert et al., the set up of Fig. (3.1) is also motivated, to an almost equal extent, by the Marinatto and Weber's idea of playing a quantum version of a matrix game.

3.3 ELW Games

I would like to mention here a quote from the paper by Eisert, Wilkens and Lewenstein: "There are several reasons why quantizing games may be interesting. First, classical game theory is a well established discipline of applied mathematics which has found numerous applications in economy, psychology, ecology and biology. Since it is based on probability to a large extend, there is a fundamental interest in generalizing this theory to the domain of quantum probabilities. Second, if the **Selfish Genes** are reality, we may speculate that games of survival are being played already on the molecular level where quantum mechanics dictates the rules. Third, there is an intimate connection between the theory of games and the theory of quantum communication. Indeed, whenever a player passes his decision to the other player or the game's arbiter, he in fact communicates information, which – as we live in a quantum world – is legitimate to think of as quantum information. On the other hand it has recently been transpired that eavesdropping in quantum-channel communication and optimal cloning can readily be conceived a strategic game between two or more players, the objective being to obtain as much information as possible in a given set-up. Finally, quantum mechanics may well be useful to win some specially designed zero-sum unfair games, like PQ penny flip, as was recently demonstrated by Meyer , and it may assure fairness in remote gambling ."

The Quantum Prisoner's Dilemma as we know it was proposed by ELW. Here I try to show another aspect of the same.

Nash Equilibria

The average payoff for a quantum game is given by:

$$\$_A = rP_{CC} + pP_{DD} + tP_{DC} + sPCD \tag{3.15}$$

In quantum prisoner's dilemma a player with access to quantum strategies can atleast do as better as a classical player.
Assuming Bob chooses $\hat{D}$, Alice's best reply would be

$$\hat{Q} = \hat{U}(0, \pi/2), \hat{Q} = C = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \tag{3.16}$$

while assuming Bob plays $\hat{C}$ Alice's best strategy would be defection $\hat{D}$. Thus, there is no dominant strategy left for Alice. The game being symmetric, the same holds for Bob, i.e., $\hat{D} \otimes \hat{D}$ is no longer an equilibrium in dominant strategies.
However, with the disappearance of the equilibrium $\hat{D} \otimes \hat{D}$ a new Nash equilibrium $\hat{Q} \otimes \hat{Q}$ has emerged with pay-off $\$_A(\hat{Q}, \hat{Q}) = \$_B(\hat{Q}, \hat{Q}) = 3$.Indeed,

$$\$_A(\hat{U}(\phi,]theta), \hat{Q}) = \cos\theta/2^2(3\sin\phi^2 + \cos\phi^2) \leq 3 \tag{3.17}$$

for all $\theta \in [0, \pi]$ and $\phi \in [0, \pi/2]$ and analogously $\$_B(\hat{Q}, \hat{U}_B) \leq \$_B(\hat{Q}, \hat{Q})$ for all $\hat{U}_B \in S$ such that no player can gain from unilaterally deviating from $\hat{Q} \otimes \hat{Q}$.

However if we want to see the real difference brought out by opening the set of quantum strategies we allow one player to be quantum and restrict the other to just classical moves. Using Eisert's **miracle** move:

$$\hat{M} = \hat{U}(\pi/2, \pi/2, 0) = \frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{3.18}$$

that yields a payoff of $< \$_B > = 3 + 2\sin\theta$ for Bob while for Alice with only $< \$_A > = 1/2$ (1 -$\sin\theta$) Here the dilemma is removed in favour of the quantum player.
ELW have shown the same in the following plots by choosing a certain parametrization such that the strategies $\hat{U}_A$ and $\hat{U}_B$ each depend on a single parameter $t \in [-1, 1]$ only: we set $\hat{U}_A = \hat{U}(t\pi, 0)$ for $t \in [0, 1]$ and $\hat{U}_A = \hat{U}(0, -t\pi/2)$ for $t \in [-1, 0)$ (same for Bob). Defection $\hat{D}$ corresponds to the value $t = 1$, cooperation $\hat{C}$ to $t = 0$, and $\hat{Q}$ is represented by $t = -1$.

Figure 3.2: Alice's pay-off in a separable game.Here $D \otimes D$ is the Nash equilibrium with a payoff of 1 for each player.

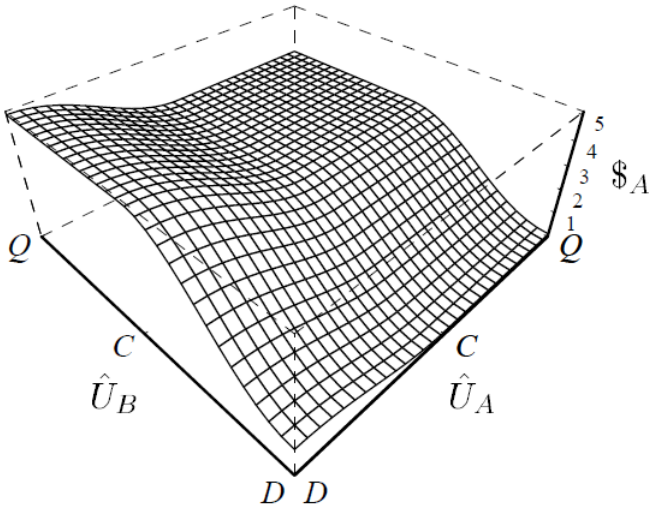
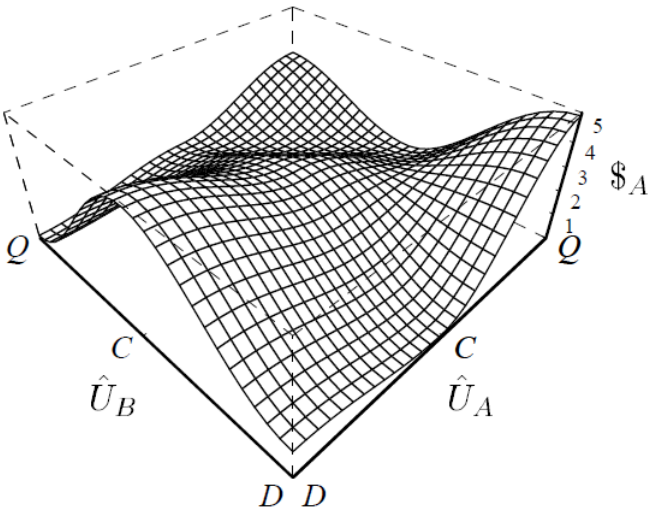


Figure 3.3: Alices pay-off for a maximally entangled game.Here $Q \otimes Q$ is the Nash equilibrium with a payoff of 3 for each player.



Bibliography

[1] Osborne, M. J. An Introduction to Game Theory Oxford University Press (2002).

[2] Studies in the Theory of Quantum Games ,Azhar Iqbal (2004) arXiv:quant-ph/0503176

[3] Quantum Parrondo's Games, Adrian P. Flitney, Joseph Ng, Derek Abbott (2002) arXiv:quant-ph/0201037

[4] Variations on the Theme of the Greenberger-Horne-Zeilinger Proof, L. Vaidman (1999) arXiv:quant-ph/9808022

[5] J S Bell,Speakable and Unspeakable in Quantum Mechanics

[6] Quantum Games and Quantum Strategies, J. Eisert, M. Wilkens, M. Lewenstein (1999) arXiv:/quant-ph/9806088

[7] Nicolas Brunner and Noah Linden, Connection Between Bell Nonlocality and Bayesian Game Theory, Nature, 2013